11.5.5.2 Default arguments

In C++, one can assign default values to the formal parameters of a function prototype. The Default arguments allows to omit some arguments when calling the function.

When calling a function,

- For any missing arguments, complier uses the values in default arguments for the called function.
- The default value is given in the form of variable initialization.

Example : **void defaultvalue(int n1=10, n2=100);**

- The default arguments facilitate the function call statement with partial or no arguments.

Example : **defaultvalue(x,y);**

defaultvalue(200,150);

defaultvalue(150);

defaultvalue(x,150);

- The default values can be included in the function prototype from right to left, i.e., we cannot have a default value for an argument in between the argument list.

Example : **void defaultvalue(int n1=10, n2);**//invalid prototype

void defaultvalue(int n1, n2 = 10);//valid prototype

11.5.5.3 Constant Arguments

The constant variable can be declared using **const** keyword. The **const** keyword makes variable value stable. The constant variable should be initialized while declaring. The **const** modifier enables to assign an initial value to a variable that cannot be changed later inside the body of the function.

Syntax :

<returntype><functionname> (const <datatype variable=value>)

Example:

- int minimum(const int a=10);
- float area(const float pi=3.14, int r=5);

Program 11.16

```
#include <iostream>
using namespace std;
double area(const double r,const double pi=3.14)
{
        return(pi*r*r);
}
int main ()
{
        double rad,res;
        cout<<"\nEnter Radius :";
        cin>>rad;
        res=area(rad);
        cout << "\nThe Area of Circle ="<<res;
        return 0;
}
```

Output:

```
Enter Radius :5
The Area of Circle =78.5
```


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